

14.2 A Geometric View of SVD



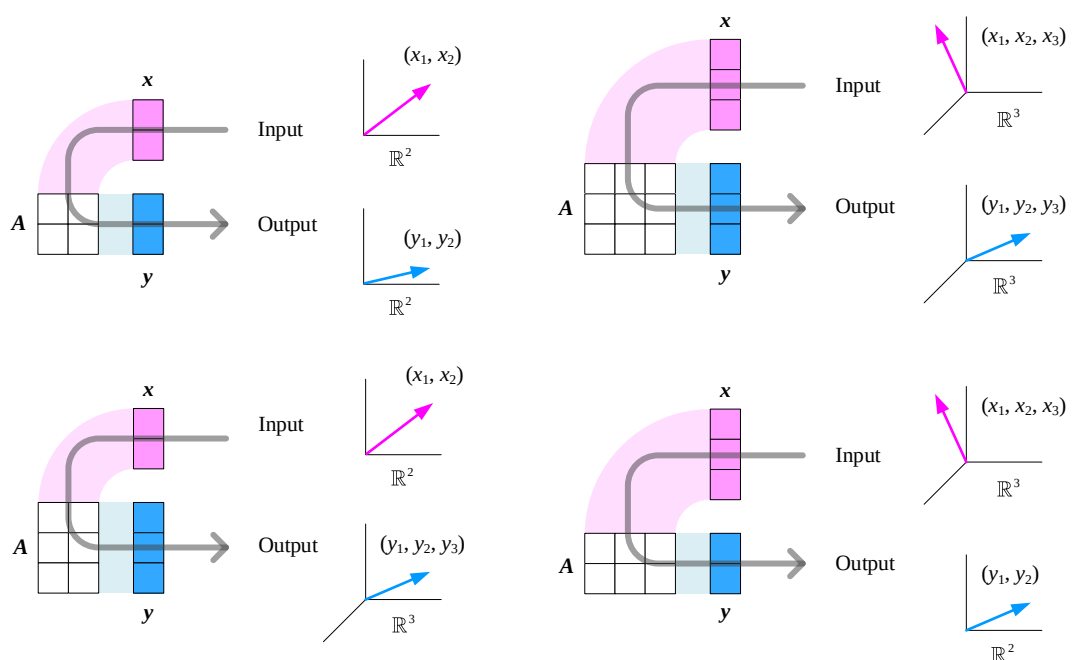
The core skills you'll master in this section:

- ▶ SVD decomposes a complex linear transformation into three steps: "rotation → scaling → rotation."
- ▶ Rotation may occur in different spaces.
- ▶ Dimension changes may occur during scaling.
- ▶ The SVD of a symmetric matrix is equivalent to its spectral decomposition.
- ▶ The area/volume change of a geometric transformation is governed by the singular values.
- ▶ A singular value of 0 means dimensionality reduction — vectors are projected onto a lower-dimensional space, reflecting a loss of rank.

To help you better understand singular value decomposition, this section introduces the SVD of four differently shaped matrices from a geometric perspective. Before studying this section, please review Chapter 2, Section 5, as well as the various geometric operations in Chapter 8 of this book.

Four Differently Shaped Matrices

Specifically, we'll analyze, in Figure 1, how four differently shaped matrices — 2×2 , 3×3 , 3×2 , and 2×3 — act on vectors.



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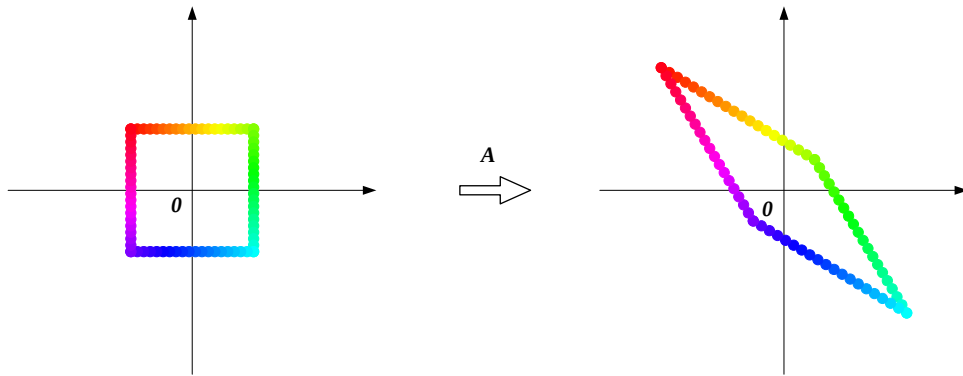
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Figure 1. Linear Mapping for Matrices of Four Different Shapes: $\mathbf{A}$ 2×2 Square Matrix

Let's first look at 2×2 square matrices. Given the following matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

Figure 2 shows the geometric operation that the 2×2 square matrix performs on the 2D column vector $\mathbf{x}$. We find that this geometric transformation cannot be reduced to a single type of operation, but is instead an ordered composition of several basic geometric operations (such as rotation, shearing, and scaling), forming a composite linear transformation. Singular value decomposition can help us understand this composite linear transformation.

Figure 2. 2×2 Symmetric Matrix $\mathbf{A}$'s Linear Transformation

Matrix $\mathbf{A}$'s SVD result is

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

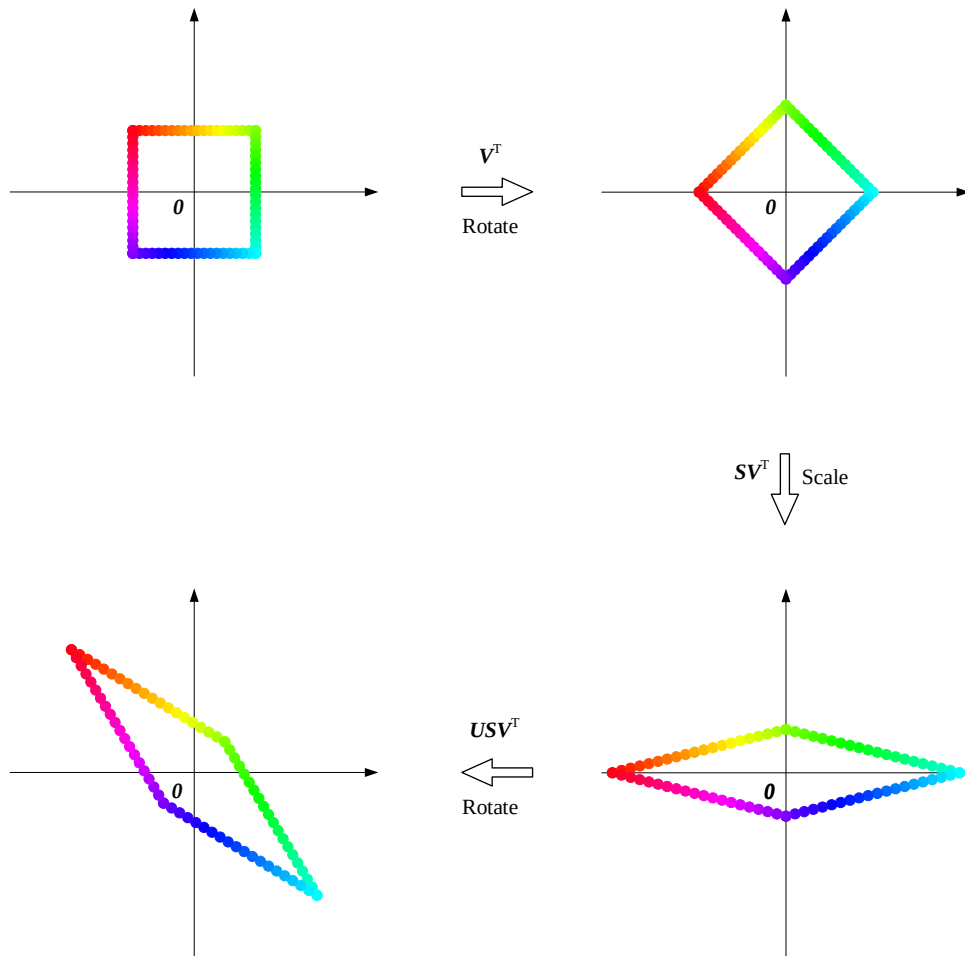
$\mathbf{A}\mathbf{x}$ can be written as $\mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{x}$, and the corresponding order of geometric operations is $\mathbf{V}^T$ (rotation) $\rightarrow \mathbf{S}$ (scaling) $\rightarrow \mathbf{U}$ (rotation), as shown specifically in Figure 3.

Since, in the equation above, matrix $\mathbf{A}$ is a real symmetric matrix, its SVD is consistent with the spectral decomposition result.

As shown, $\mathbf{V}^T$ and $\mathbf{U}$ respectively correspond to rotation operations that do not change the area. The diagonal matrix $\mathbf{S}$ determines the scaling of the shape.

Computing the determinant gives 1, meaning the shape's area is unchanged — that is, $\det(\mathbf{A}) = \det(\mathbf{S})$. Since the determinant is positive, the shape is also not flipped.

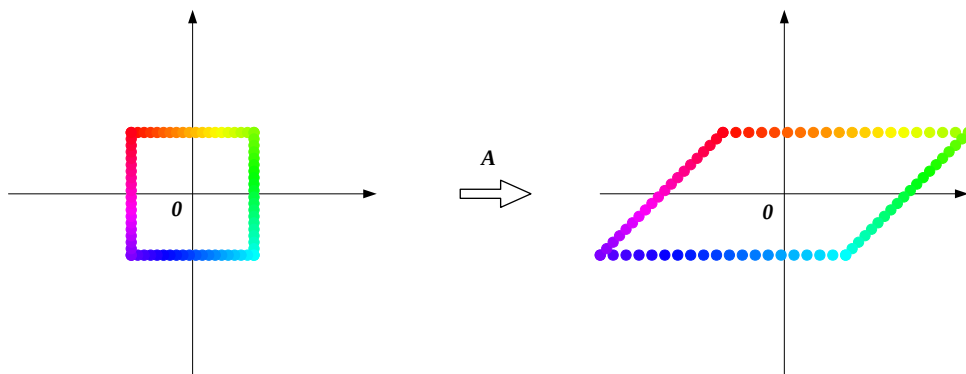
In particular, since $\mathbf{U}$ and $\mathbf{V}^T$ are inverse transformations of each other, their rotation angles are exactly opposite.

Figure 3. 2×2 Symmetric Matrix A 's SVD Result USV^T 's Corresponding Linear Transformation

Let's look at another 2×2 non-symmetric matrix A , specifically as follows

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Figure 4 shows the linear transformation produced by the 2×2 non-symmetric matrix A .

Figure 4. 2×2 Non-Symmetric Matrix A 's Linear Transformation

In the equation above, for matrix A , the SVD result is

$$A = USV^T = \underbrace{\begin{bmatrix} 0.973 & -0.229 \\ 0.229 & 0.973 \end{bmatrix}}_U \underbrace{\begin{bmatrix} 2.288 & 0 \\ 0 & 0.874 \end{bmatrix}}_S \underbrace{\begin{bmatrix} 0.851 & 0.526 \\ -0.526 & 0.851 \end{bmatrix}}_{V^T} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

⚠ Note that the values in the equation above keep only three decimal places.

As shown in Figure 5, in the equation above, V^T 's geometric operation is a rotation of the plane about the origin. S scales up along the horizontal axis and scales down along the vertical axis. U 's operation is likewise a rotation of the plane about the origin.

Since (in the equation above) A is not a symmetric matrix, U 's rotation is not the inverse of V^T 's operation.

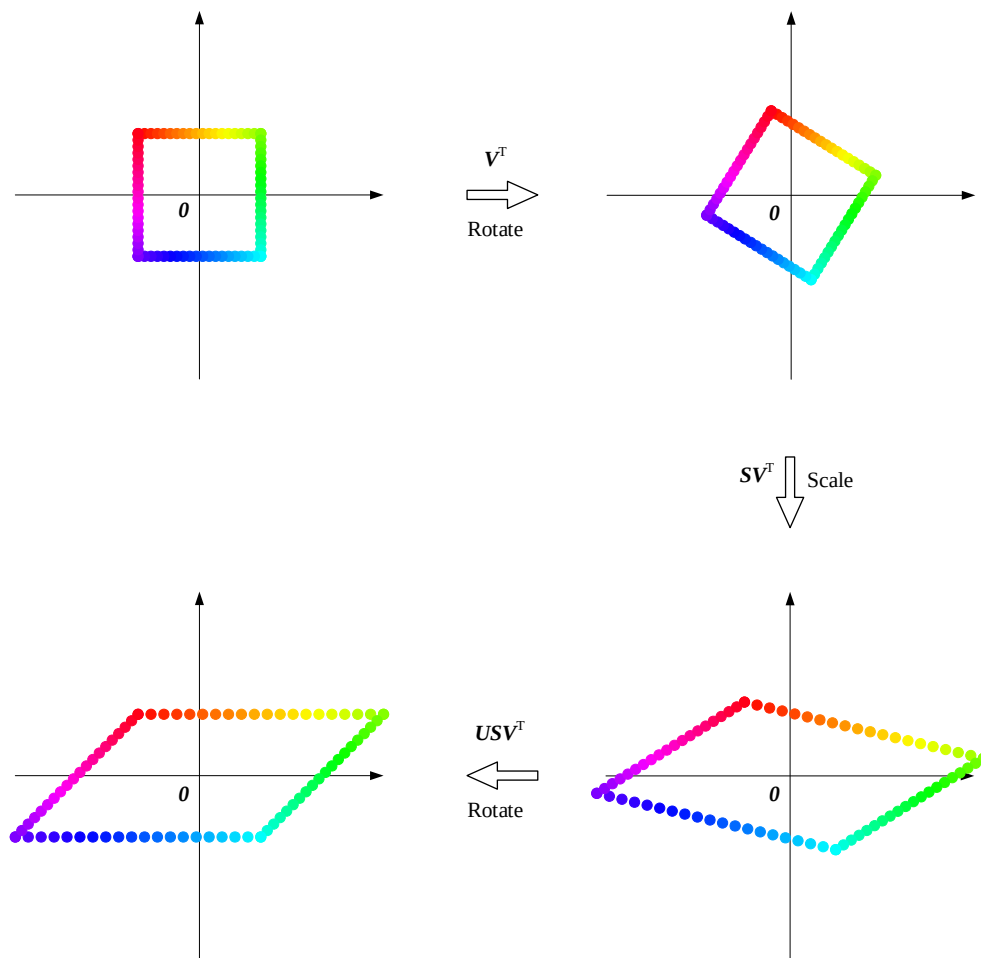


Figure 5. 2×2 Non-Symmetric Matrix A 's SVD Result USV^T 's Corresponding Linear Transformation

3×3 Square Matrix

Next, let's discuss the singular value decomposition of 3×3 square matrices.

Given the following 3×3 symmetric square matrix A

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

In the equation above, matrix $\mathbf{A}$ is full rank, which means its column vectors are not collinear, and also means its determinant is nonzero.

Figure 6 shows, in the equation above, the 3×3 symmetric square matrix $\mathbf{A}$'s geometric transformation.

 Please use Laplace expansion to compute, from the equation above, the determinant of matrix $\mathbf{A}$.

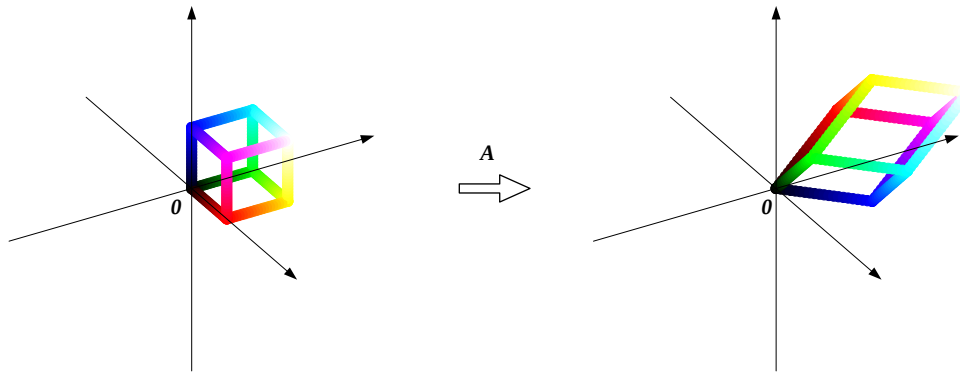


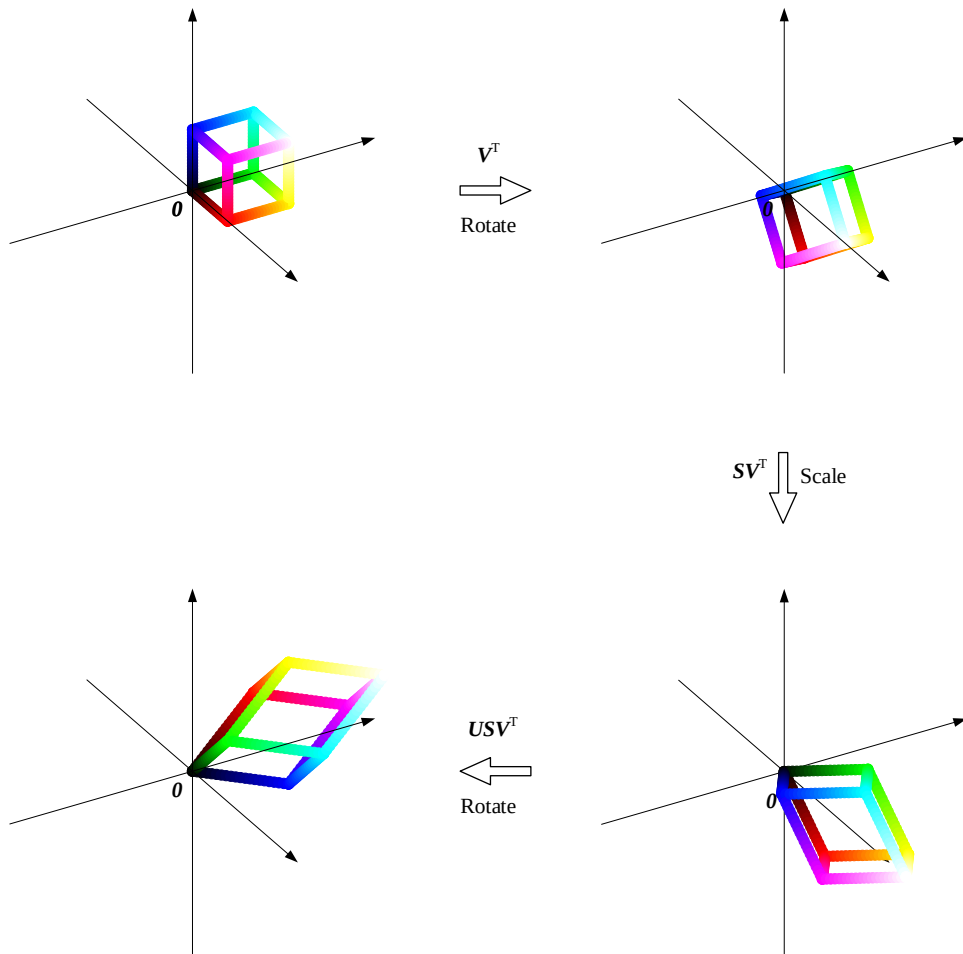
Figure 6. 3×3 Matrix $\mathbf{A}$ (Full Rank) – Linear Transformation

In the equation above, the 3×3 square matrix $\mathbf{A}$'s SVD is

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T = \underbrace{\begin{bmatrix} \sqrt{3}/3 & 0 & \sqrt{6}/3 \\ \sqrt{3}/3 & -\sqrt{2}/2 & -\sqrt{6}/6 \\ \sqrt{3}/3 & \sqrt{2}/2 & -\sqrt{6}/6 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} \sqrt{3}/3 & \sqrt{3}/3 & \sqrt{3}/3 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ -\sqrt{6}/3 & \sqrt{6}/6 & \sqrt{6}/6 \end{bmatrix}}_{\mathbf{V}^T} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

As shown in Figure 7, in the equation above, $\mathbf{V}$'s geometric operation is a rotation in 3D space. $\mathbf{S}$ scales in 3D space. $\mathbf{U}$'s operation is likewise a rotation in 3D space.

It's easy to compute $\det(\mathbf{S}) = 2$. Since $\mathbf{A}$ is a square matrix, $\det(\mathbf{A}) = \det(\mathbf{S})$, so $\det(\mathbf{A})$ is also 2.

Figure 7. 3×3 Symmetric Matrix $\mathbf{A}$ (Full Rank) — SVD Result $\mathbf{USV}^T$'s Corresponding Linear Transformation

Let's look at another 3×3 square matrix that is not full rank, $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

We easily find that $\mathbf{a}_1 - \mathbf{a}_2 = \mathbf{a}_3$, which shows that the square matrix $\mathbf{A}$'s column vectors are linearly dependent, so its determinant is clearly 0. As shown in Figure 8, $\mathbf{A}$'s linear transformation causes a reduction in dimension!

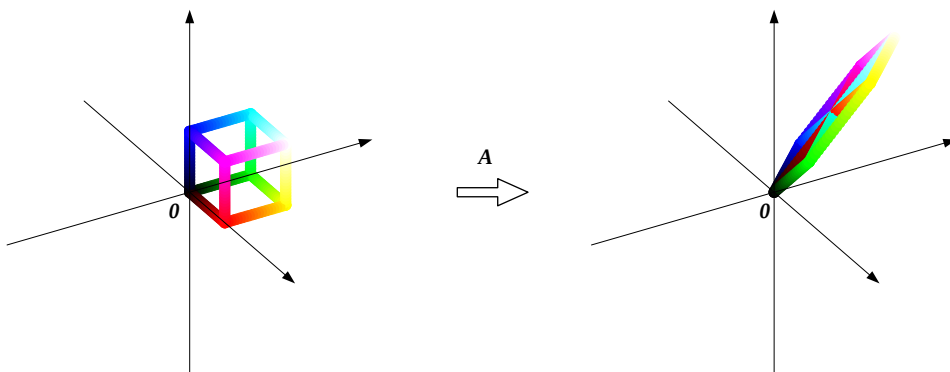


Figure 8. 3×3 Matrix A (Not Full Rank) — Linear Transformation

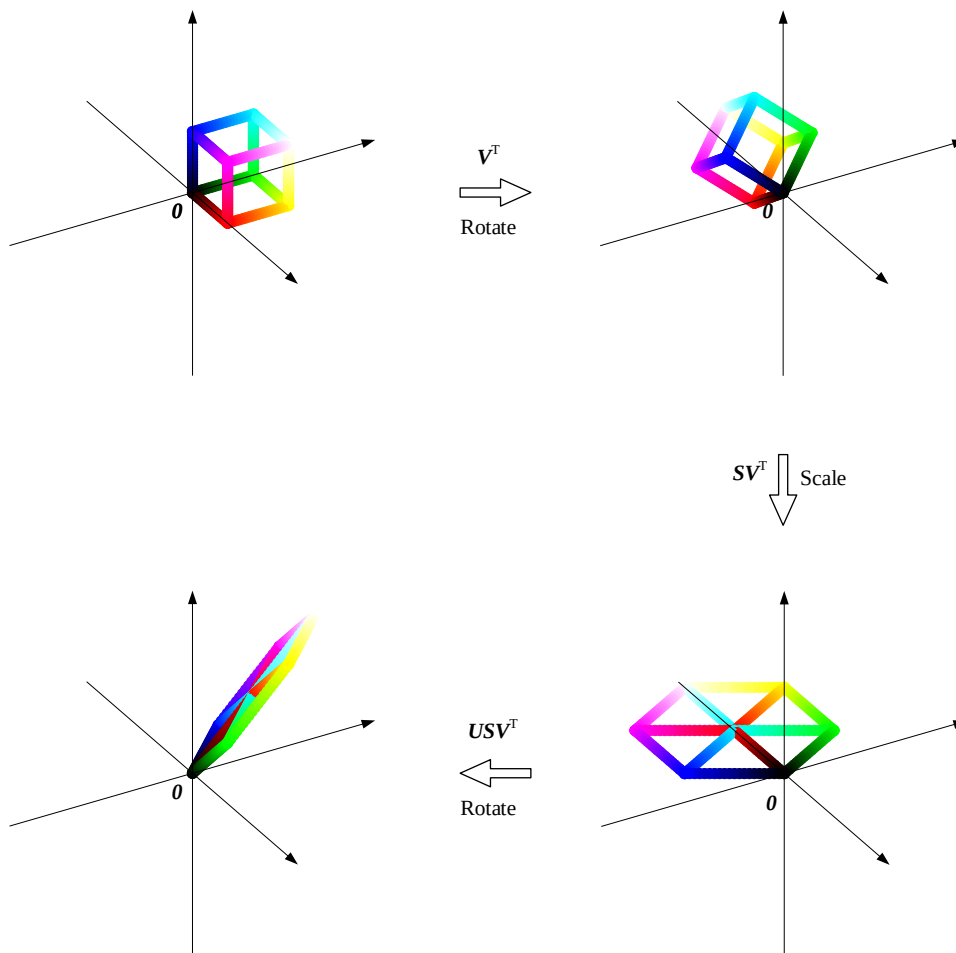
In the equation above, the 3×3 square matrix A 's SVD

$$A = USV^T = \underbrace{\begin{bmatrix} 0 & \sqrt{6}/3 & -\sqrt{3}/3 \\ \sqrt{2}/2 & -\sqrt{6}/6 & -\sqrt{3}/3 \\ \sqrt{2}/2 & \sqrt{6}/6 & \sqrt{3}/3 \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sqrt{3} & & \\ & \sqrt{3} & \\ & & 0 \end{bmatrix}}_S \underbrace{\begin{bmatrix} \sqrt{6}/3 & \sqrt{6}/6 & \sqrt{6}/6 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ -\sqrt{3}/3 & \sqrt{3}/3 & \sqrt{3}/3 \end{bmatrix}}_{V^T} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

We find that one of the singular values is 0.

Note that, in the equation above, A has rank 2; U and V are full rank, while S also has rank 2.

As shown in Figure 9, in the equation above, V^T 's geometric operation is a rotation in 3D space. S performs "scaling + dimensionality reduction" in 3D space. U 's operation is likewise a rotation in 3D space.

Figure 9. 3×3 Non-Symmetric Matrix A (Not Full Rank) — SVD Result USV^T 's Corresponding Linear Transformation

3×2 Tall-and-Thin Matrix

Let's look at the singular value decomposition of a 3×2 tall-and-thin matrix.

Given the following 3×2 tall-and-thin matrix A

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$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Clearly, in the equation above, matrix $\mathbf{A}$ is full column rank. $\mathbf{Ax} = \mathbf{y}$'s corresponding geometric operation is shown in Figure 10. The column vector $\mathbf{x}$ lies in a plane, while $\mathbf{y}$ lies in 3D space; therefore, $\mathbf{Ax} = \mathbf{y}$ achieves a "dimension increase"!

Looking at Figure 10, we can easily see that although the vector has expanded from 2D to 3D, it still lies within a single plane.

➔ This point was covered in detail in Chapter 2, Section 5, when discussing the geometric perspective of matrix multiplication — please review it.

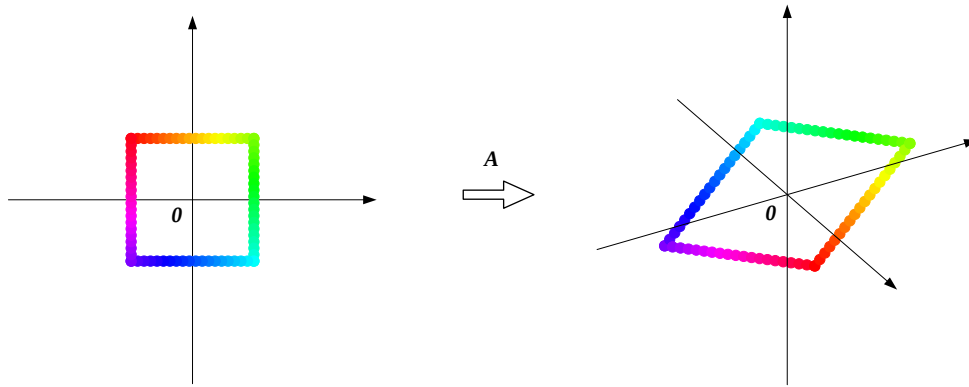


Figure 10. 3×2 Tall-and-Thin Matrix $\mathbf{A}$ (Full Column Rank) — Linear Transformation

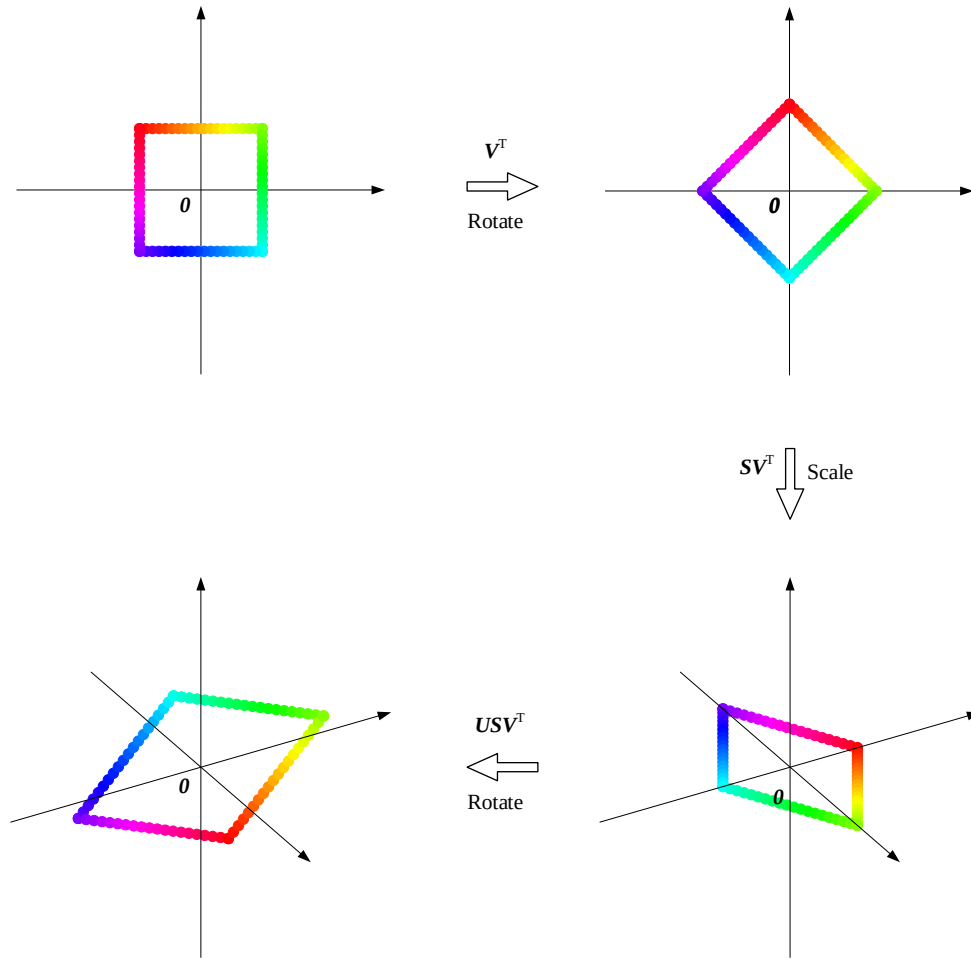
In the equation above, this matrix $\mathbf{A}$'s full SVD is

$$\mathbf{A} = \mathbf{USV}^T = \underbrace{\begin{bmatrix} \sqrt{6}/6 & \sqrt{2}/2 & \sqrt{3}/3 \\ \sqrt{6}/3 & 0 & -\sqrt{3}/3 \\ \sqrt{6}/6 & -\sqrt{2}/2 & \sqrt{3}/3 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Note that, in the equation above, $\mathbf{A}$ has rank 2; $\mathbf{U}$ and $\mathbf{V}$ are full rank, while $\mathbf{S}$ also has rank 2.

As shown in Figure 11, $\mathbf{V}$ rotates in the plane (about the origin). $\mathbf{S}$ performs "scaling + dimension increase". $\mathbf{U}$'s operation is a rotation in 3D space.

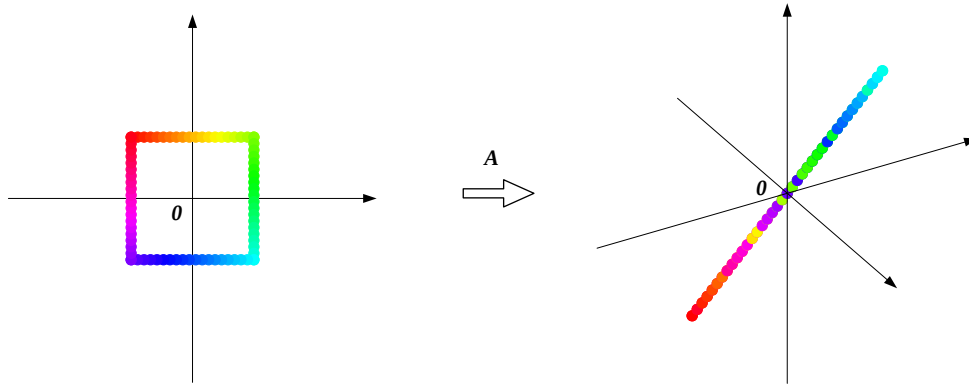
As an analogy, $\mathbf{S}$ in the equation above is like placing a photo (2D) on a table (3D).

Figure 11. 3×2 Tall-and-Thin Matrix $\mathbf{A}$ (Full Column Rank) — SVD Result $\mathbf{USV}^T$'s Corresponding Linear Transformation

Given the following 3×2 tall-and-thin matrix that is not full column rank $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$$

Figure 12 shows, from the equation above, the linear transformation. We find that all the vectors fall on a single line.

Figure 12. 3×2 Tall-and-Thin Matrix $\mathbf{A}$ (Not Full Column Rank) — Linear Transformation

In the equation above, matrix $\mathbf{A}$'s full SVD is

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ \sqrt{2}/2 & 0 & -\sqrt{2}/2 \\ \sqrt{2}/2 & 0 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^T} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$$

As shown in Figure 13, $\mathbf{V}$ rotates in the plane (about the origin). $\mathbf{S}$ performs "scaling + dimension increase". $\mathbf{U}$'s operation is a rotation in 3D space.

Since $\mathbf{S}$ has one scaling factor equal to 0, that direction is essentially a dimensionality reduction — this is why in Figure 13 all the scattered points fall on the same line.

Note that, in the equation above, $\mathbf{A}$ has rank 1; $\mathbf{U}$ and $\mathbf{V}$ are full rank, while $\mathbf{S}$ also has rank 1.

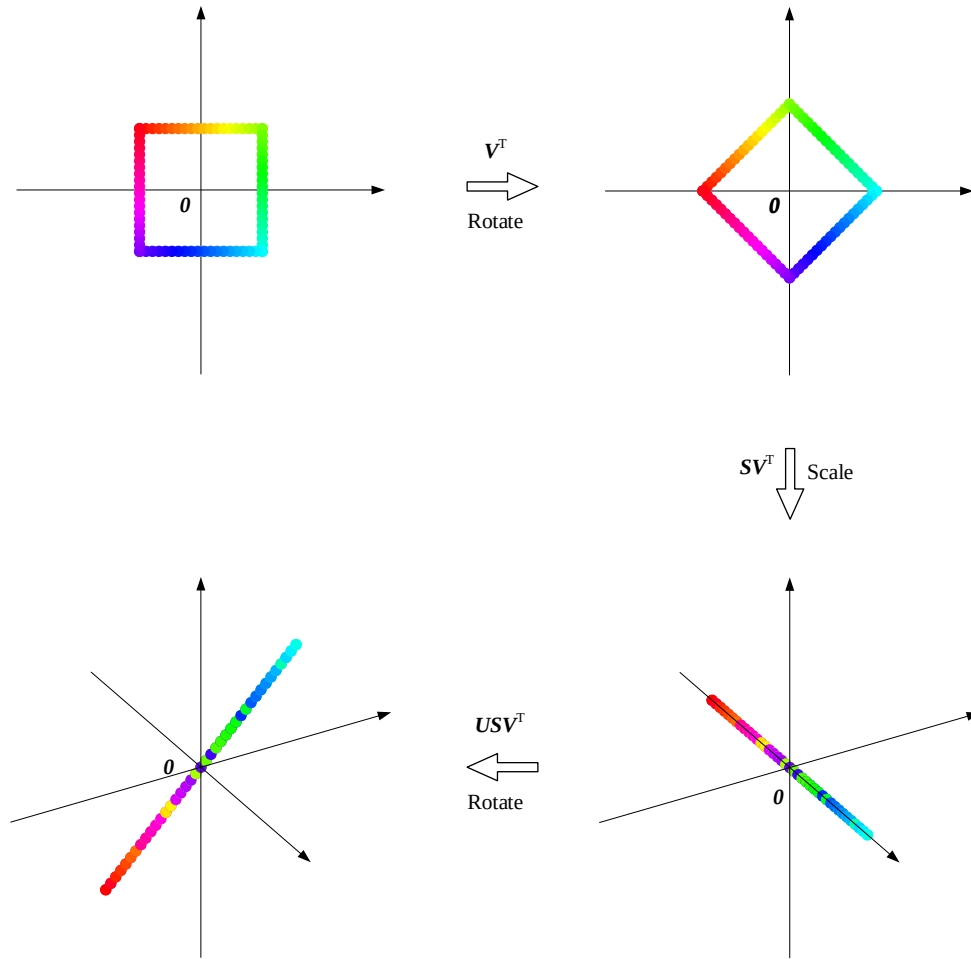


Figure 13. 3×2 Tall-and-Thin Matrix A (Full Column Rank) – SVD Result USV^T 's Corresponding Linear Transformation

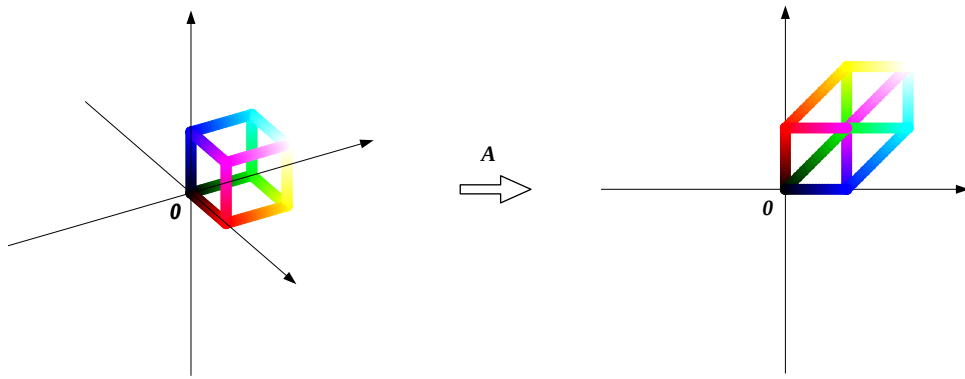
2×3 Flat Matrix

Finally in this section, let's look at 2×3 flat matrices.

Given the following row-full-rank 2×3 flat matrix A

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$Ax = y$'s corresponding geometric operation is shown in Figure 14. The column vector x lies in 3D space, while y lies in a plane; therefore, $Ax = y$ achieves "dimensionality reduction". We can see in Figure 14 that the unit cube is indeed compressed onto a plane.

Figure 14. 2×3 Flat Matrix A (Row Full Rank) — Linear Transformation

In the equation above, this matrix A 's full SVD is

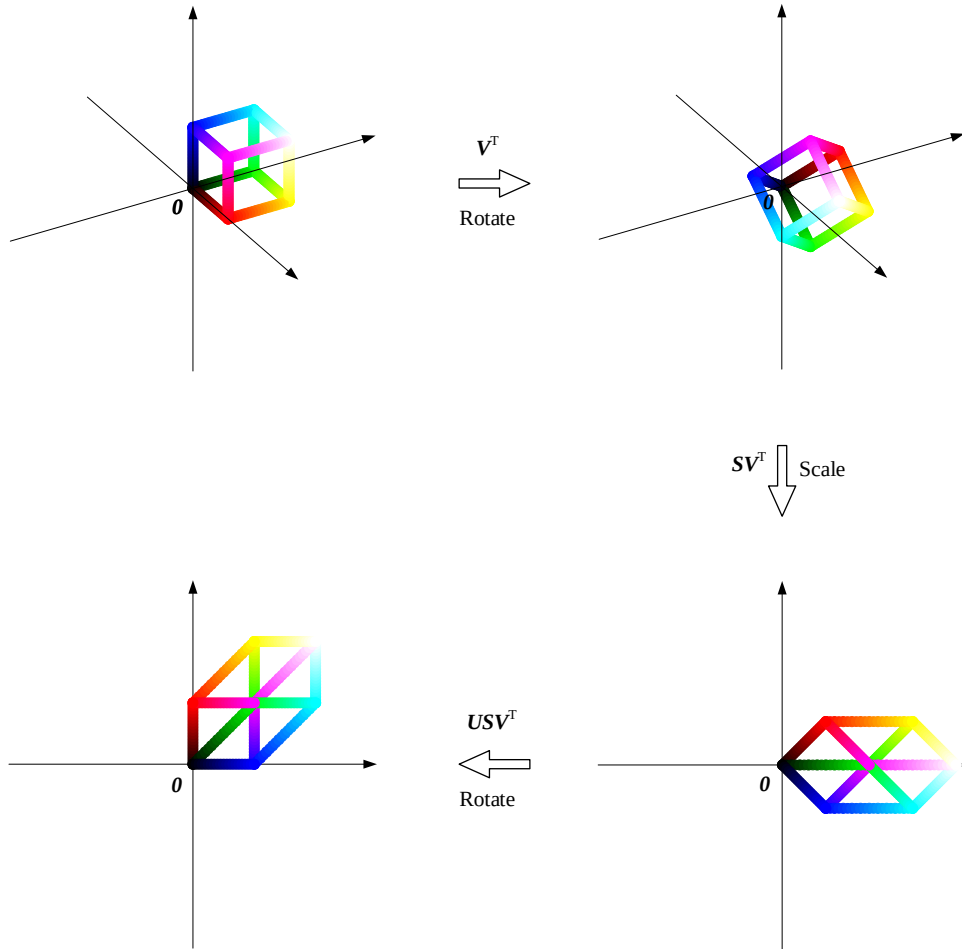
$$A = \underbrace{USV^T}_{\substack{U \\ S \\ V^T}} = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \underbrace{\begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_S \underbrace{\begin{bmatrix} \sqrt{6}/6 & \sqrt{6}/3 & \sqrt{6}/6 \\ \sqrt{2}/2 & 0 & -\sqrt{2}/2 \\ \sqrt{3}/3 & -\sqrt{3}/3 & \sqrt{3}/3 \end{bmatrix}}_{V^T} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

As shown in Figure 15, V^T first rotates the unit cube in 3D space.

S makes the vectors undergo "scaling + dimensionality reduction". This is like taking a photo of a 3D geometric shape — the image becomes two-dimensional.

Finally, U rotates the planar geometric shape about the origin.

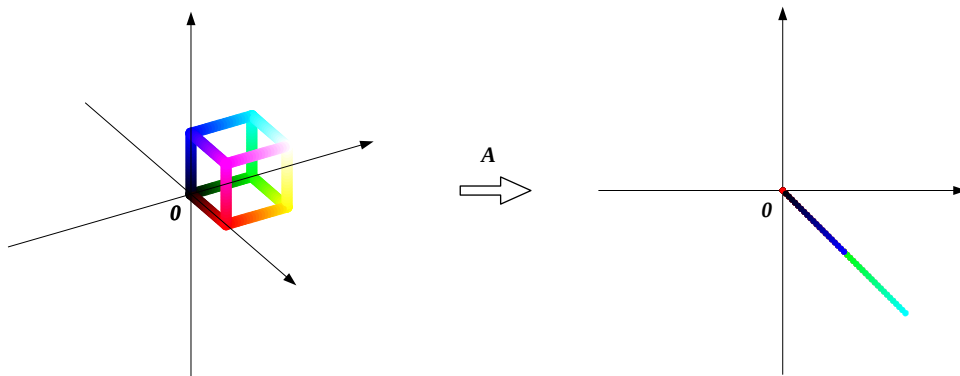
Note that, in the equation above, A has rank 2; U and V are full rank, while S also has rank 2.

Figure 15. 2×3 Flat Matrix A (Row Full Rank) – SVD Result USV^T 's Corresponding Linear Transformation

Let's look at the last example in this section. Given the following 2×3 flat matrix A

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

It's easy to see that the matrix in the equation above is not full row rank; its corresponding linear transformation is shown in Figure 17.

Figure 16. 2×3 Flat Matrix A (Not Full Row Rank) – Linear Transformation

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In the equation above, matrix A 's full SVD is

$$A = USV^T = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_U \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_S \underbrace{\begin{bmatrix} 0 & \sqrt{2}/2 & \sqrt{2}/2 \\ 1 & 0 & 0 \\ 0 & -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{V^T} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

In the equation above, the sequential geometric operations corresponding to the decomposed matrices are shown in Figure 17.

? Please analyze Figure 17 on your own.

Note that, in the equation above, A has rank 1; U and V are full rank, while S also has rank 1.

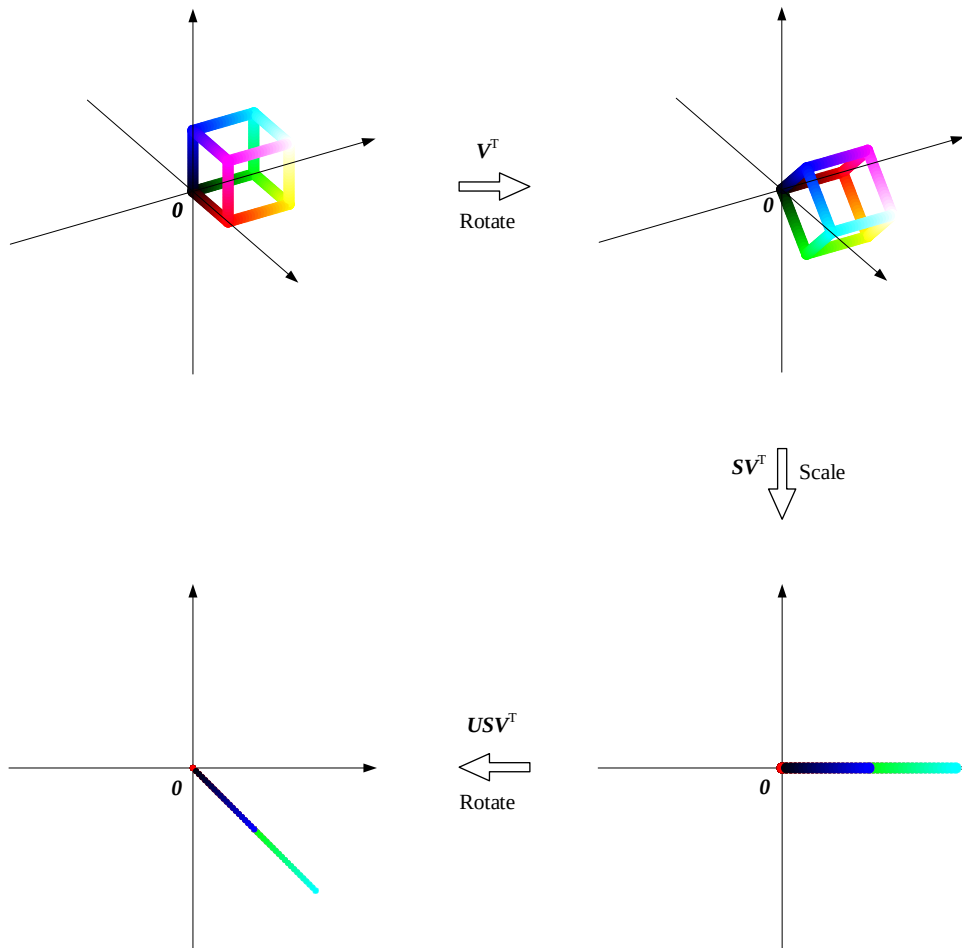


Figure 17. 2×3 Flat Matrix A (Not Full Row Rank) — SVD Result USV^T 's Corresponding Linear Transformation



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please hand-calculate the singular value decomposition for all the examples in this section, and verify the results with Python code.

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